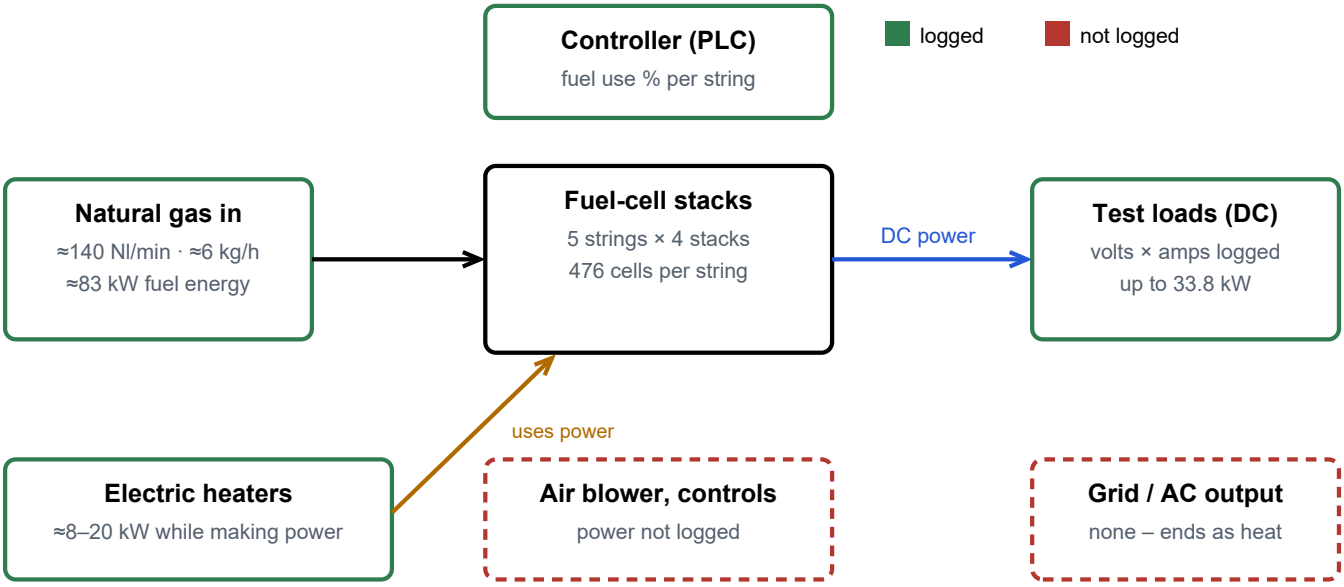
SOFC POC data · 39 days · 835 columns checked · 2026-11-04

Does the POC data hold any efficiency figures?

Contents: 1. Is it logged? · 2. Calculated efficiency · 3. Tag check and next steps · 4. At rated power

No column is called “efficiency”. But the data has everything needed to work it out, and the controller already logs one closely related number: **fuel utilisation**, the share of the gas that the cells actually turn into electricity.

That matters because efficiency is the headline number for any fuel-cell plant. We can now put a figure on it for the POC runs instead of leaving it blank.



Where the energy goes. Green boxes have numbers in the data; red dashed boxes do not. The plant sends its power to test loads that burn it as heat, so there is no grid or AC output to measure.

What we found

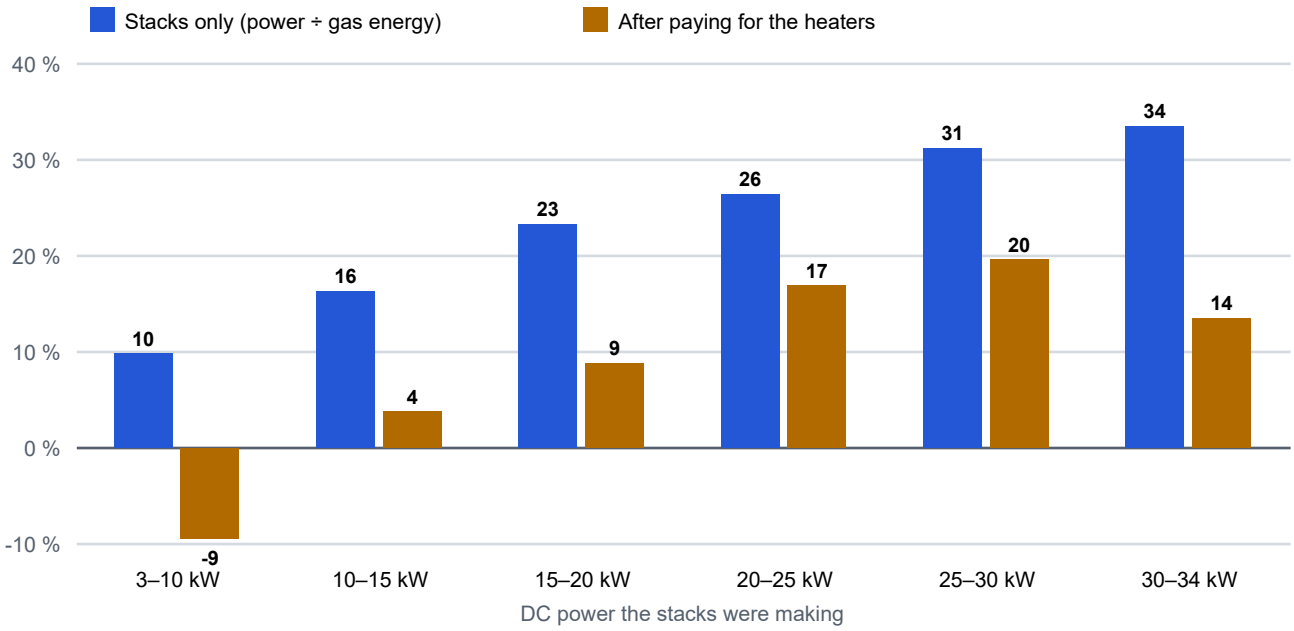
Item	Status	What it means
Efficiency tag	NOT LOGGED	No efficiency column on any of the 39 days. The heater's “effective set-point” is a heater setting, not an efficiency.
Fuel utilisation	LOGGED	One value per string from the controller; matches our own calculation almost perfectly.
Calculated efficiency	CAN BE WORKED OUT	Power out ÷ gas energy in: about 21 % typical, up to 35 % at the highest power reached.
Other devices	PARTLY	Heater power is logged and can be subtracted. Blower power is not, so a true plant figure is still out of reach.

If you remember one thing: efficiency is not recorded, but it can be calculated: about one fifth of the gas energy becomes electricity on a typical POC run, and about one third at the highest power reached.

Efficiency rises with power

At low power only about 10 % of the gas energy becomes electricity. At the highest power reached (≈34 kW) it is about 33 %.

Think of a car stuck in first gear: it burns fuel just to keep the engine warm, so little goes into moving. The stacks work the same way. A lot of the gas keeps them hot; the more current we draw, the bigger the share that becomes useful power.



Median of every second the plant made more than 3 kW (19, 24, 25, 30 March 2026). Blue = stacks only. Amber = after taking away the electric heaters' power. Below 10 kW the heaters use more than the stacks make.

How it is worked out

Efficiency = power out (volts × amps of the 5 strings) ÷ gas energy in (gas flow × 50 MJ per kg of methane). **Fuel utilisation** = current drawn compared with the most current the gas could give; ours and the controller's agree within 3 %. **After heaters** subtracts heater power first; blower and control power are not logged, so the true plant figure is lower.

The numbers

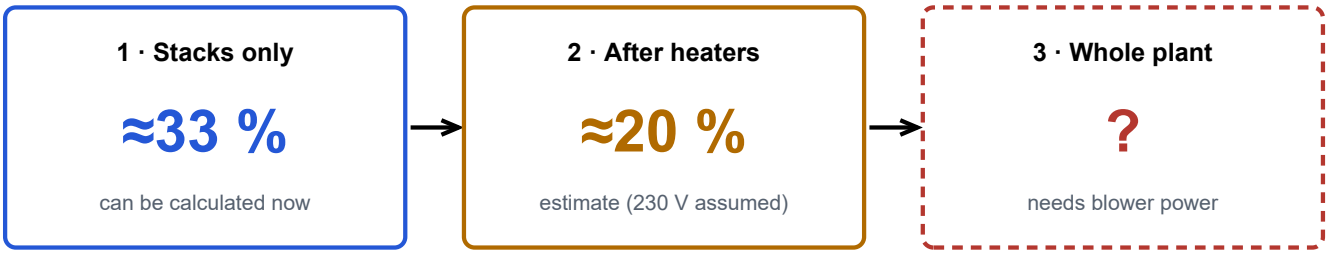
Power band	Minutes	Gas energy kW	Efficiency	Heaters kW	After heaters	Fuel use
3–10 kW	76	66	9.8 %	14.4	-9.4 %	10.2 %
10–15 kW	35	78	16.3 %	9.8	3.8 %	17.5 %
15–20 kW	44	72	23.3 %	10.5	8.8 %	25.5 %
20–25 kW	46	83	26.4 %	7.7	16.9 %	30.1 %
25–30 kW	27	90	31.2 %	10.0	19.6 %	36.8 %
30–34 kW	11	95	33.5 %	19.9	13.5 %	39.5 %

Fuel use stays low (10–40 %); commercial plants usually run at 60–85 %. That is the main reason efficiency is low here: the POC never pushed the stacks to rated power (5 × 14 kW = 70 kW).

If you remember one thing: efficiency here depends mostly on how hard the stacks are pushed; the POC never ran at full load, so 33 % is a part-load figure, not the plant's best.

Tag check and next steps

Only three columns are useful for efficiency: fuel utilisation per string, gas mass flow, and string volts and amps. The tags that sound most promising are empty.



Three levels of efficiency; each subtracts more of the plant's own power use. Level 2 is possible today; level 3 needs data the POC did not record.

Tag by tag

Column	What it is	Status	Note
PLC_pv_rFU_Worst_1…5	Fuel utilisation per string, %	USE IT	Matches our calculation, r = 0.999. String 4 lower (20 % vs 24 %). Junk values outside power periods.
PLC_pv_rFU_best_1…5	“Best” fuel utilisation	EMPTY	Zero whenever the plant makes power.
PLC_pv_rM_CH4	Gas mass flow, kg/h	USE IT	Equals gas flow FIRC 1,4 converted as pure methane.
EIRC_4x_VoltageMon × CurrentMon	String DC power	USE IT	Clean. EIRC_4x_Actual_Power has gaps and 3 × 10^{37} junk values.
PLC_pv_rSystemOutPower	System output power	EMPTY	Always 0 on all 39 days.
PLC_pv_rRejectionRatio	Unclear (likely gas recycle)	EMPTY	0 while making power.
pi_EIRC_2_01_…_EffSetpoint	Air-heater effective set-point	NOT EFFICIENCY	The heater's power setting.
pi_IIR_*, EIRC_2_01 Power	Heater currents, air-heater power	PARTLY	Currents need a voltage (230 V assumed).
—	Blower, controls, AC output	NOT LOGGED	Needed for a true plant efficiency.

Things to keep in mind

Power was made on only four days (19, 24, 25, 30 March 2026), about 4 hours above 3 kW. Gas energy assumes pure methane (50 MJ/kg); real natural gas differs by a few percent, shifting every efficiency figure by the same amount. The controller's fuel-utilisation reads 3.3 % higher than ours, probably for the same reason.

Decisions for you

A · Gas energy value. Keep pure methane (50 MJ/kg), or ask LG for the real gas analysis?
B · Add to the pipeline. Add efficiency and fuel use as calculated columns in the notebooks and digital twin?
C · Ask the plant for blower and control-cabinet power, for a whole-plant figure?
D · Heater voltage. Confirm 230 V for the heater currents (changes only the “after heaters” figure).

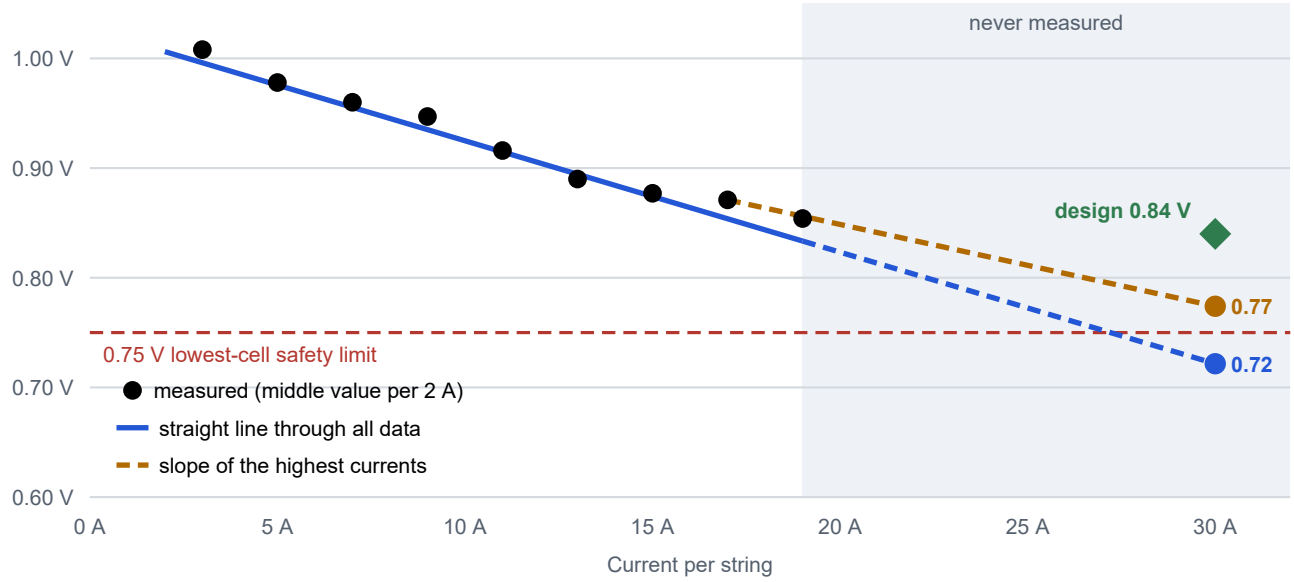
Calculation script: data_analysis/scripts/efficiency_check.py

If you remember one thing: use the per-string fuel-utilisation tags and gas flow with volts × amps; ignore “System Out Power” and “Rejection Ratio”, which are empty.

At rated current: about 52–55 kW and 52–56 % efficiency

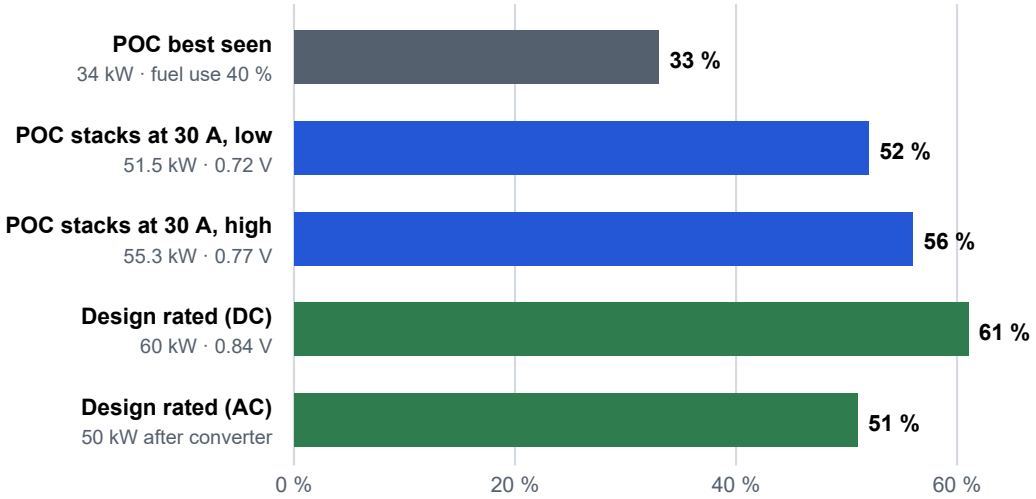
The design says 60 kW at 0.84 V per cell. Extending what the POC stacks actually did, they would give **0.72–0.77 V per cell** at the rated 30 A. That means **51–55 kW** and about **52–56 %** efficiency, if the gas is set so 75 % of it is used.

The design rated voltage is 400 V per string: 100 V per stack, or 0.84 V per cell. The POC stacks already drop to that voltage at about 17–20 A, roughly 60 % of the rated current. Running them harder pulls the voltage lower, like a battery that sags under a heavy load.



Average cell voltage against string current, from every second with 2 A or more on the 4 power days. Beyond 19 A nothing was measured, so the dashed lines are estimates. The design point is missed by 0.07–0.12 V.

Efficiency in each case



Gross DC efficiency with the gas set for 75 % fuel use: 7.1 kg/h, about 166 Nl/min, almost exactly the most the POC ever fed (165 Nl/min). AC uses the design 50 kW AC from 60 kW DC.

Why efficiency almost doubles

Efficiency = fuel use × cell voltage ÷ 1.04 V. In the POC the gas was already turned up close to the rated amount, but only about half the rated current was drawn, so fuel use stayed at 40 %. At 30 A on the same gas, fuel use rises to about 75 % while the voltage only falls from 0.84 to about 0.75 V.

Fuel use	Gas kg/h	Gas Nl/min	Gas energy kW	POC low 0.72 V	POC high 0.77 V	Design 0.84 V
70 %	7.6	177	106	49 %	52 %	57 %
75 %	7.1	166	99	52 %	56 %	61 %
80 %	6.7	155	93	56 %	60 %	65 %
85 %	6.3	146	87	59 %	63 %	69 %

Fuel use is set by the operator through the gas. Real plants usually stay at or below 80–85 %, because the last few percent of fuel starves the cells and damages them.

Limits to this estimate

Safety limit: the controller stops at 0.75 V on the weakest cell, which sits below the average. String 3 would average 0.67 V at 30 A, so in practice the POC stacks would stop before 30 A. **Hotter stacks:** at rated power the stacks heat themselves more and lose resistance, which pushes the voltage toward the higher estimate. **Plant's own use:** these are gross DC figures; heaters, blower and the converter take their share (design: 50 kW AC from 60 kW DC).

If you remember one thing: at the rated 30 A the POC stacks would give roughly 52–55 kW (not the design 60 kW) at about 52–56 % DC efficiency. The jump from 33 % comes from using the gas the plant already feeds.